

Quiz (Habanero)

- Q1.** What is the slope of a line parallel to $y = 2x + 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** If two lines have the same slope, what can be concluded about them?
(A) Perpendicular
(B) Parallel
(C) Intersecting
(D) Skew
(E) None
- Q3.** A line has a slope of -2 and passes through $(1, 3)$. What is the equation of a line parallel to it?
(A) $y = -2x + 1$
(B) $y = -2x + 5$
(C) $y = -2x - 1$
(D) $y = 2x + 1$
(E) $y = 2x - 1$
- Q4.** In a given ecosystem, the relationship between temperature (T) and rainfall (R) can be modeled by the equation $R = 2T + 5$. If a parallel line to this relationship has a y -intercept of 11, what is its equation?
(A) $R = 2T + 11$
(B) $R = 2T - 1$
(C) $R = T + 5$
(D) $R = T - 3$
(E) $R = 3T + 2$
- Q5.** If $y = 3x - 2$ and $y = 3x + 1$ are two parallel lines, what is the distance between them?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q6.** The graphs of $y = x + 2$ and $y = x - 1$ are examples of which type of lines?
(A) Perpendicular
(B) Parallel
(C) Intersecting
(D) Skew
(E) None
- Q7.** A line passes through $(2, 5)$ and is parallel to $y = x - 3$. What is its equation?
(A) $y = x + 1$
(B) $y = x + 3$
(C) $y = x - 1$
(D) $y = x - 3$
(E) $y = x + 5$
- Q8.** If $y = 2x - 1$ and $y = 2x + 3$ are parallel, what is the relationship between their y -intercepts?
(A) Equal
(B) Different
(C) Increasing
(D) Decreasing
(E) Constant

Open Ended Questions (FRQ)

- Q9.** A weather forecast models the relationship between temperature and atmospheric pressure using the equation $P = 0.5T + 10$. Find the equation of a line parallel to this relationship that passes through the point $(20, 15)$. Show all work.
- Q10.** In an ecosystem, the relationship between the amount of rainfall (R) and the amount of plant growth (G) is modeled by $G = 2R + 5$. If a parallel line to this relationship has a y -intercept of 8, find the equation of this line and explain its significance in the context of the ecosystem. Use proper math notation (\$) *forequations*.

Chef's Tips

- Ensure students understand the concept of parallel lines and their applications in real-world scenarios.
- Remind students to show all work for free-response questions.
- Emphasize the importance of using proper math notation.